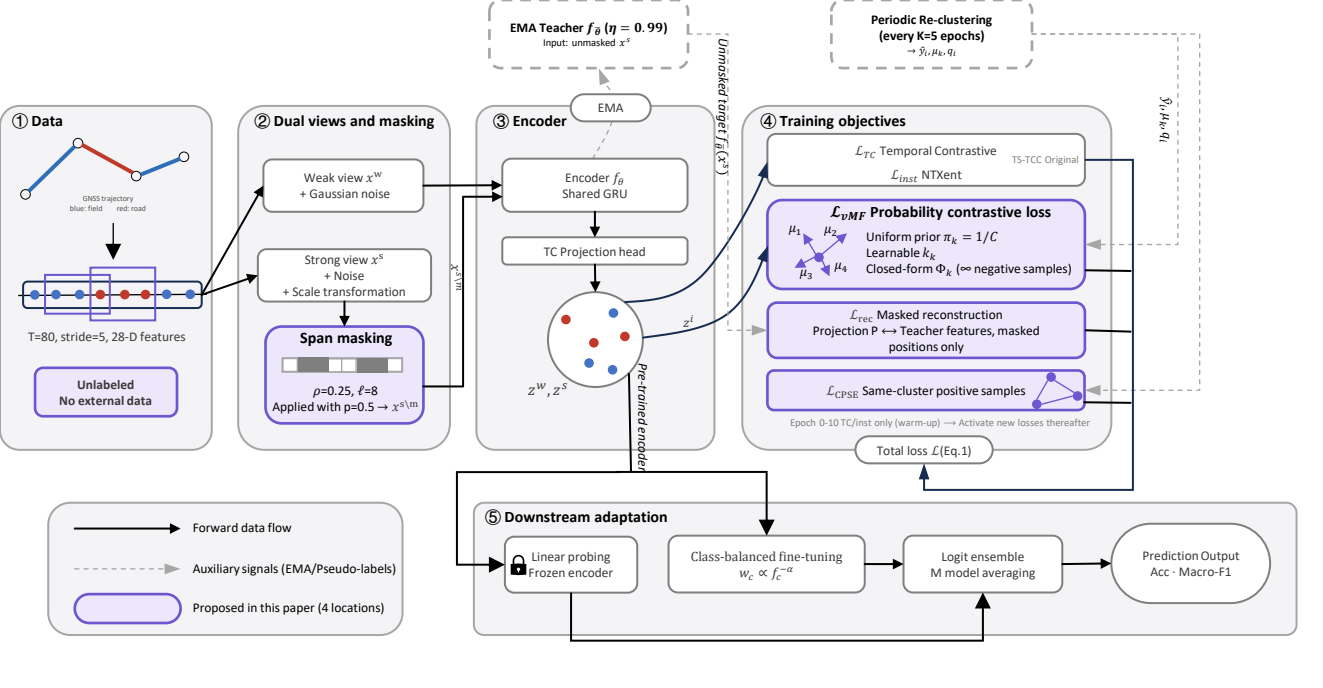

EMA Teacher $f_{\bar{\theta}}$ ($\eta = 0.99$)
Input: unmasked x^s
Periodic Re-clustering
(every K=5 epochs)
→ $\hat{y}_i, \mu_k, q_i$
$\hat{y}_i, \mu_k, q_i$
Unmasked target $f_{\bar{\theta}}(x^s)$
① Data
GNSS trajectory
blue: field red: road
T=80, stride=5, 28-D features
Unlabeled
No external data
② Dual views and masking
Weak view x^w
+ Gaussian noise
Strong view x^s
+ Noise
+ Scale transformation
Span masking
ρ=0.25, ℓ=8
Applied with p=0.5 → $x^{s\setminus m}$
$x^w, x^{s\setminus m}$
③ Encoder
EMA
Encoder f_θ
Shared GRU
TC Projection head
z^i
z^w, z^s
Pre-trained encoder
④ Training objectives
$\mathcal{L}_{TC}$ Temporal Contrastive
TS-TCC Original
$\mathcal{L}_{inst}$ NTXent
$\mathcal{L}_{vMF}$ Probability contrastive loss
μ_1 μ_2 μ_3 μ_4
Uniform prior $\pi_k = 1/C$
Learnable k_k
Closed-form Φ_k (∞ negative samples)
$\mathcal{L}_{\text{rec}}$ Masked reconstruction
Projection P ↔ Teacher features, masked positions only
$\mathcal{L}_{\text{CPSE}}$ Same-cluster positive samples
Epoch 0-10 TC/inst only (warm-up) → Activate new losses thereafter
Total loss $\mathcal{L}$(Eq.1)
⑤ Downstream adaptation
Linear probing
Frozen encoder
Class-balanced fine-tuning
$w_c \propto f_c^{-\alpha}$
Logit ensemble
M model averaging
Prediction Output
Acc · Macro-F1
Forward data flow
Auxiliary signals (EMA/Pseudo-labels)
Proposed in this paper (4 locations)